PropIQ — Real Estate AI

AI-Powered House Price Prediction for the USA

Project Proposal Report

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Course: Algorithms for Data Science

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GitHub: github.com/samira2a04-AI/house-price-prediction
Dashboard: https://house-price-prediction-j5j39rbb97z9vf4r2nur7s.streamlit.app/

1. Title

PropIQ — AI-Powered House Price Prediction for the USA

2. Problem Statement

Accurately estimating residential property values is a persistent challenge in the real estate industry. Buyers, sellers, investors, and financial institutions rely on trustworthy price estimates to make well-informed decisions. Traditional appraisal methods are time-consuming, expensive, and prone to subjective bias. With the explosion of publicly available real estate listing data, there is a clear opportunity to leverage machine learning to produce faster, more objective, and scalable property valuations.

This project addresses the problem of predicting the sale price of residential properties across the United States using a large-scale dataset of over 483,000 listings. The core challenge lies in capturing the complex, non-linear interactions among a wide range of features — including structural attributes (bedrooms, bathrooms, square footage), locational intelligence (ZIP code, state, city), and engineered composite indicators (luxury score, density metrics, price-per-square-foot ratios) — to generate accurate and generalizable predictions.

3. Objectives

The primary objectives of this project are:

- ✦ Build a robust end-to-end machine learning pipeline capable of predicting U.S. residential house prices with high accuracy.
- ✦ Perform thorough exploratory data analysis (EDA) to understand data distributions, correlations, and geographic patterns.
- ✦ Design and implement a comprehensive data preprocessing workflow, including missing value imputation, outlier detection and treatment, and categorical encoding.
- ✦ Engineer meaningful new features — such as luxury scores, bedroom-to-bathroom ratios, and ZIP-code-level density metrics — to boost predictive power.
- ✦ Train and compare multiple gradient-boosted tree models (XGBoost and LightGBM) using robust cross-validation strategies.
- ✦ Deploy an interactive Streamlit web application that allows end users to receive instant price estimates by entering property details.
- ✦ Achieve an R^2 score above 0.80 and a Mean Absolute Error (MAE) below $100,000 on the held-out test set.

4. Dataset Description

The project utilizes a rich USA housing dataset containing over 483,000 real estate listings sourced from Kaggle. The dataset is publicly available at: kaggle.com/datasets/ahmedshahriarsakib/usa-real-estate-dataset and provides wide geographic coverage, supporting 99.85% of U.S. ZIP codes.

4.1 Key Features

- Structural attributes: number of bedrooms, bathrooms, house size (sq ft), lot size, number of stories, and garage presence.
- Location data: ZIP code, city, state, and geographic coordinates (latitude, longitude).
- Listing metadata: property type, listing status, and sale price (target variable).
- Engineered features: luxury score, price-per-square-foot, bedroom-to-bathroom ratio, and ZIP-level density metrics.

4.2 Data Challenges

- Missing values across multiple columns, handled through hierarchical imputation strategies (ZIP-level → city-level → state-level medians).
- Significant price outliers requiring careful capping and filtering to avoid model distortion.
- High-cardinality categorical features (e.g., ZIP codes, cities) requiring target encoding or embedding strategies.

5. Methodology

The project follows a structured data science pipeline consisting of the following phases:

5.1 Data Collection & Exploration

Initial data loading and exploratory data analysis (EDA) to understand variable distributions, missing data patterns, geographic spread, and correlations with the target variable (house price).

5.2 Data Preprocessing

- Missing value imputation using a hierarchical approach: ZIP-code median → city median → state median → global median.
- Outlier detection using IQR-based methods and domain-knowledge thresholds.
- Encoding of categorical features using target encoding for high-cardinality columns and one-hot encoding for low-cardinality ones.
- Log transformation of the target variable to normalize the right-skewed price distribution.

5.3 Feature Engineering

- Luxury Score: a composite metric combining house size, lot size, bedroom count, bathroom count, and garage presence.
- Ratios: bedroom-to-bathroom ratio, house-size-to-lot-size ratio.

- Location Intelligence: ZIP-level average price, listing density, and distance-to-urban-center proxies.

5.4 Model Training & Validation

Two primary models are trained and compared:

- XGBoost (eXtreme Gradient Boosting): a high-performance gradient boosting framework optimized for speed and accuracy.
- LightGBM (Light Gradient Boosting Machine): a fast, memory-efficient alternative particularly suited for large tabular datasets.

Model validation is conducted using Group K-Fold Cross-Validation, where groups are defined by ZIP code, ensuring that listings from the same geographic area do not appear in both training and validation folds. This prevents data leakage and provides a realistic estimate of generalization performance.

5.5 Hyperparameter Tuning

Grid search and randomized search are applied to optimize key hyperparameters, including learning rate, maximum tree depth, number of estimators, subsampling ratios, and regularization terms.

5.6 Deployment

The best-performing model is saved and integrated into a Streamlit web application, providing an interactive dark-luxury UI for end users to input property attributes and receive instant price predictions with confidence ranges.

6. Evaluation Metrics

The following metrics are used to evaluate model performance:

- R^2 Score (Coefficient of Determination): measures the proportion of variance in house prices explained by the model. A score of 1.0 indicates perfect prediction.
- Mean Absolute Error (MAE): the average absolute difference between predicted and actual prices, interpretable directly in US dollars.
- Root Mean Squared Error (RMSE): penalizes large errors more heavily than MAE, useful for detecting systematic over- or under-predictions.
- Mean Absolute Percentage Error (MAPE): expresses prediction error as a percentage of the actual price, enabling comparison across different price ranges.

6.1 Achieved Results

Metric	Value	Description
R^2 Score	0.82	Excellent model fit

MAE	~$92,000	Mean Absolute Error
Dataset Size	483,000+ listings	Training records
ZIP Coverage	99.85%	U.S. ZIP codes covered

7. Tools & Technologies

Category	Tool / Library
Programming Language	Python 3.9+
ML Libraries	XGBoost, LightGBM, Scikit-learn
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly
Web Application	Streamlit
Development Environment	Jupyter Notebook
Version Control	Git & GitHub

8. Expected Outcomes

Upon successful completion of this project, the following outcomes are expected:

- A fully trained and validated machine learning model capable of predicting U.S. residential house prices with an R^2 score of at least 0.82 and an MAE of approximately $92,000.
- A clean, reproducible Jupyter Notebook documenting the complete data science pipeline from raw data ingestion through model evaluation.
- A deployed Streamlit web application (PropIQ) with a modern, intuitive interface enabling non-technical users to obtain property valuations by entering basic house and location details.
- Insights into the most influential features driving house prices across U.S. regions, delivered through feature importance plots and geographic visualizations.
- A reusable preprocessing and feature engineering pipeline that can be adapted to other real estate datasets or geographic regions.
- Comprehensive project documentation on GitHub, including code, notebooks, requirements, and a detailed README.

Live App: house-price-prediction-j5j39rbb97z9vf4r2nur7s.streamlit.app

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